

WASP-39b

R-1214

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_220531 · created 2026-07-31T08:05:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459730.7841 +/- 0.00075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.011
Transit depth (Rp/Rs)^2	2.73 +/- 0.36 [%]
Orbital Inclination (inc)	87.8 +/- 0.56
Airmass coefficient 1 (a1)	1.326 +/- 0.011
Airmass coefficient 2 (a2)	-0.1433 +/- 0.0072
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	None
Optimal Aperture	3.56
Optimal Annulus	9.08
Transit Duration (day)	0.1178 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.011 vs 0.1457 (deviation 1.75σ)
Duration fit vs published	0.1178 d vs 0.12 d (deviation 0.61σ)
Mid-time O–C	1.8 min
Depth SNR	15.0
Residual scatter	0.81 %

Light curve

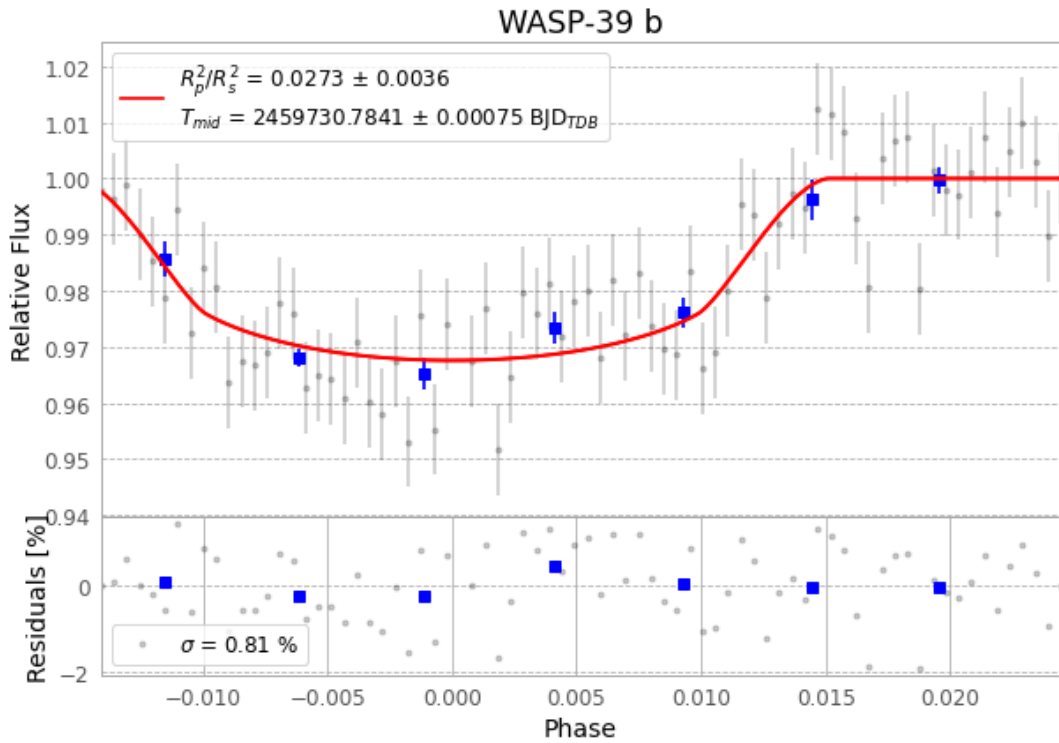


Figure 1 — FinalLightCurve_WASP-39 b_2022-05-30.png (SHA-256 5a81323d7723b646...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [396, 97], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 4440755f4faae79e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-39220531051813.FITS → WASP-39220531090313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-220531.log (7cb357b4960c...), FinalLightCurve_WASP-39 b_2022-05-30.png (5a81323d7723...), FinalLightCurve_WASP-39 b_2022-05-30.csv (e78dd0847dc9...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:05Z.